

The CRR Binomial Option Pricing Model

An analysis of Accuracy, Convergence and Stability Using Python

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Abstract

In finance, the binomial options pricing model provides a generalizable numerical method for the valuation of options. Essentially, the model uses a "discrete-time" (lattice based) model of the varying price over time of the underlying financial instrument, addressing cases where the closed-form Black-Scholes formula is wanting. The binomial model was first proposed by William Sharpe in the 1978 and was formalized by Cox, Ross and Rubinstein (CR) in 1979 and by Rendleman and Bartter in that same year. This paper will calculate the value of the range of options under the CRR model using Python. It will then investigate the pricing accuracy and convergence of the algorithm. We will then investigate other methods to increase the efficiency, convergence, and stability of the method.

Keywords: Binomial Option Pricing, Convergence, Black-Scholes

Introduction

In 1979, John Cox, along with Stephen Ross and Mark Rubinstein, made a major breakthrough in simplifying the way options were priced. Following the groundbreaking work of (Black & Scholes, 1973) in the field of option pricing, (Cox, Ross, & Rubinstein, 1979) established the foundation for the numerical method that is known as the binomial option pricing model by improving upon the work proposed by Sharpe. This was a simple discrete-time model for valuing options as opposed to the continuous-time approach followed by the Black-Scholes Model (BSM). This significantly simplified the way options were priced and paved the way for much headway in this field, most notably work by (Jarrow & Rudd, 1982), (Rendleman Jr & Bartter, 1979), (Tian, 1999).

1 The Binomial Lattice Model

The basic idea behind the model is quite simple. It involves dividing the life of the option into a large number of small time intervals of length δt . It assumes that in each time interval the price of the underlying asset moves from its initial value of S_0 to one of two new values, S_u and S_d (Hull, 2009).

This is repeated a total of n times until the strike date is reached and a total of 2^n possible terminal values of the underlying are determined. From this the intrinsic value of the option is determined at each node and then discounted at rate r to move backward through the tree to determine the value of the option at each node, denoted by $V_{j,n}$. This is repeated until the last node point $V_{0,0}$ is reached. This will be the price of the option.

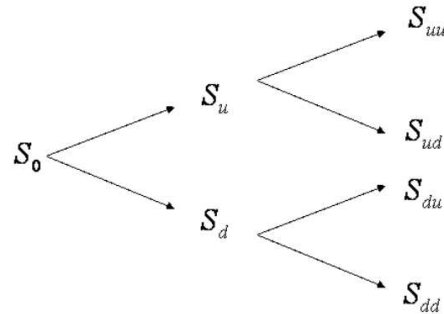


Figure 1: A Two-Step Binomial Model

When we want to move from S_0 all the way through the tree and back to determine its price, there are a few things to consider:

1. By how much does the price of S_0 move when it moves up to S_u ? It is important to note that $S_u = uS_0$, i.e. we need to know what the value of u is. Similarly, $S_d = dS_0$.
2. What is the probability that the price of underlying will move from S_0 to either S_u or S_d ? In the former case, the probability is denoted by p , with the latter being $(1 - p)$.
3. Finally, at what rate are we discounting?

The answer to 1 and 2 depends on the models chosen and will be discussed as we move on. As for the discount rate consider (Merton, 1974). Robert Merton proposed setting up a risk-less portfolio consisting of the option and the underlying asset (stock). He the argued that the return of the portfolio over a short period of time must be the risk-free return. This is denoted by r .

2 The Cox-Ross-Rubinstein (CRR) Model

2.1 Asset Price Process

In a risk neutral continuous time model, the asset price S is assumed to follow a log-normal process. Over a period δt , the asset price $S_{\delta t}$ can be expressed in terms of $S_t = S$ as

$$S_{\delta t} = S e^{(r - \frac{1}{2}\sigma^2)\delta t + \sigma\sqrt{\delta t}Z} \quad (1)$$

where σ is the volatility, and $Z \sim N(0, 1)$ a standard normal random variable.

$S_{\delta t}$ has the first moment $S e^{r\delta t}$ and the second moment $S^2 e^{(2r + \sigma^2)\delta t}$. From CR we also know the first two moments for the discrete time model, $S(pu + (1 - p)d)$ and $S^2(pu^2 + (1 - p)d^2)$. Now, by equating the first and second moments of the asset price $S_{\delta t}$ in both continuous and discrete time models, we obtain the relation:

$$pu + (1 - p)d = e^{r\delta t} \quad (2)$$

$$pu^2 + (1 - p)d^2 = e^{(2r + \sigma^2)\delta t} \quad (3)$$

There are two equations and three unknowns u , d , p . An extra condition is needed to uniquely determine a solution. Note that (2) implies $p = \frac{R - d}{u - d}$ where $R = e^{r\delta t}$

2.2 The Model

To solve (2) and (3), CR proposed the following condition:

$$ud = 1.^* \quad (4)$$

Solving (2), (3) and (4) and ignoring higher order terms, one arrives at the following key equations.

$$u = e^{\sigma\sqrt{\delta t}}, \quad (5)$$

$$d = e^{-\sigma\sqrt{\delta t}} \quad (6)$$

From here we have all we need to determine a risk-neutral value of an option! Let's consider an example using Python code (see Appendix A for the code of each pricing function). The following assumptions were used in pricing these options:

$$S_0 = \$ 100$$

$$n = 50$$

$$r = 10\%$$

$$X_1 = \$ 110$$

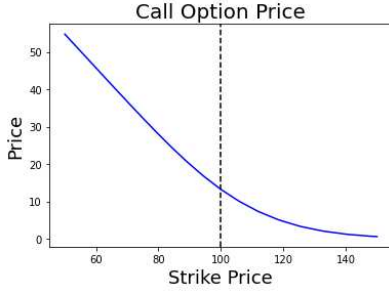
$$\sigma = 20\%$$

$$t = 1 \text{ year}$$

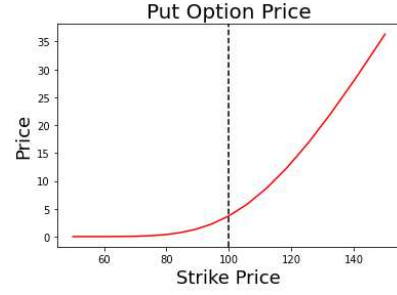
$$X_2 = \$ 90$$

Of course, my choice of $n = 50$ is arbitrary at this point and we will later see which choice of N is better. First we plot the option prices for one European Put and one European Call over a range of different strike prices:

*This condition results in symmetry of the tree



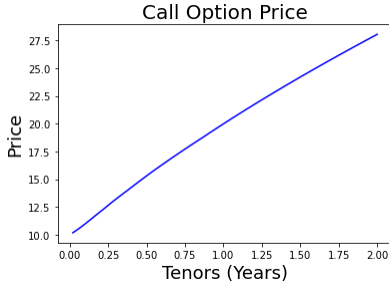
(a) Plot of Call Option Prices



(b) Plot of Put Option Prices

Figure 2: Option Prices for Various Underlying Spot Prices

As we can see, the shape of the plots matches our expectations. The dotted line in each plot represent the spot price of the underlying. Next, we plot the option prices for one European Put and one European Call over a range of different tenors[†]:



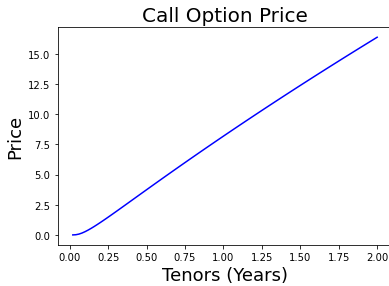
(a) Plot of Call Option Prices



(b) Plot of Put Option Prices

Figure 3: Option Prices for Various Tenors

This is where we start to see some small errors creeping in. Because we are using a discrete approximation, the curvature of the plot isn't as we would expect. Let's Consider how these exact options would plot under the Black-Scholes Pricing Model:



(a) Plot of Call Option Prices: BSM



(b) Plot of Put Option Prices: BSM

Figure 4: Option Prices for Various Tenors under BSM

[†]Note: here we use X_1 to price the put option so it is ITM, and X_2 for the call option for the same reason.

Firstly, the plots in Fig 4 are smoother and the curvature (Theta or $\frac{\partial V}{\partial t}$) around $t=0$ is more pronounced. Second, it was also not possible to plot the point $t=0$ in Fig 3 as this would mean that $\delta t = 0/n$.

2.3 Measuring Accuracy and Convergence of The CRR Model

As with all models, it is necessary to measure the accuracy and convergence of the model before it is accepted as an alternative. The question now is: What model do we use to compare the CRR model to. The natural answer would be to say the Black-Scholes Model. But we must first asses if and under what circumstances they will converge.

2.3.1 Choice of Benchmark Model

CR provides us the explanation. Consider the n -step binomial model which is represented by the following convenient formula[‡]:

$$C = \left[\sum_{j=0}^n \left(\frac{n!}{j!(n-j)!} \right) p^j (1-p)^{n-j} \max [0, u^j d^{n-j} S - X] \right] / R^n \quad (7)$$

This a complete formula[§], but with some work, it is shown that the formula can be simplified to

$$C = \left[\sum_{j=a}^n \left(\frac{n!}{j!(n-j)!} \right) p^j (1-p)^{n-j} [u^j d^{n-j} S - X] \right] / R^n \quad (8)$$

With a being a the smallest non-negative integer such that $u^j d^{n-j} S > X$. Of course, if $a > n$, the call will finish out-of-the-money even if the stock moves upward every period, so its current value must be zero. By breaking up C into two terms, we can write:

$$C = S \left[\sum_{j=a}^n \left(\frac{n!}{j!(n-j)!} \right) p^j (1-p)^{n-j} \left(\frac{u^j d^{n-j}}{R^n} \right) \right] - X R^{-n} \left[\sum_{j=a}^n \left(\frac{n!}{j!(n-j)!} \right) p^j (1-p)^{n-j} \right] \quad (9)$$

Which can be rewritten as:

$$C = S \Phi(a; n, p') - X R^{-n} \Phi(a; n, p) \quad (10)$$

Where

$$p = \frac{R - d}{u - d} \quad (11)$$

$$p' = \frac{pu}{R} \quad (12)$$

$$a = \text{the smallest non-negative integer greater than } \frac{\log(X/Sd^n)}{\log(u/d)} \quad (13)$$

$$\text{if } a > n \text{ then } C = 0 \quad (14)$$

[‡]this formula is for a European Call option

[§]here $R^n = e^{r \frac{t}{n}}$

It is then shown that the limiting case of Equation 10 as $n \rightarrow \infty$ or $\delta t \rightarrow 0$, is the Black-Scholes Option Pricing Formula[¶]:

$$C = SN(d_1) - Xe^{-rt}N(d_2) \quad (15)$$

Similar steps can be used to prove the convergence of the European Put option.

2.3.2 Experiments and Results

In this section we start to investigate the convergence of the CRR model to the BSM. We will use Python to run the experiments^{||}.

The following assumptions were used:

$S_0 = \$ 100$	$X_0 = 100$
$r = 10\%$	$X_1 = 120$
$\sigma = 20\%$	$X_2 = 80$
$t = 1 \text{ year}$	Option Type - European Call

We will measure convergence by looking at the Error, the Relative Error, l_∞ -error^{**} as well as the l_1 -error^{††} for varying number of step sizes. The formulas for these metrics are simple:

$$\begin{aligned} e &= V_{BSM} - V_{CRR}, \\ r_e &= \frac{V_{BSM} - V_{CRR}}{V_{BSM}}, \\ l_\infty &= \|e\|_\infty, \\ l_1 &= \|e\|_1 \end{aligned}$$

[¶](Thickstun, 2016) provides a more in depth proof of this result

^{||}the code is once again provided in Appendix A

^{**}denotes the maximum error in the vector of errors

^{††}denotes the sum of all the absolute errors

Experiment 1 - ATM Call

Here we consider the convergence when the option is ATM (i.e. using $X_0 = S_0$).

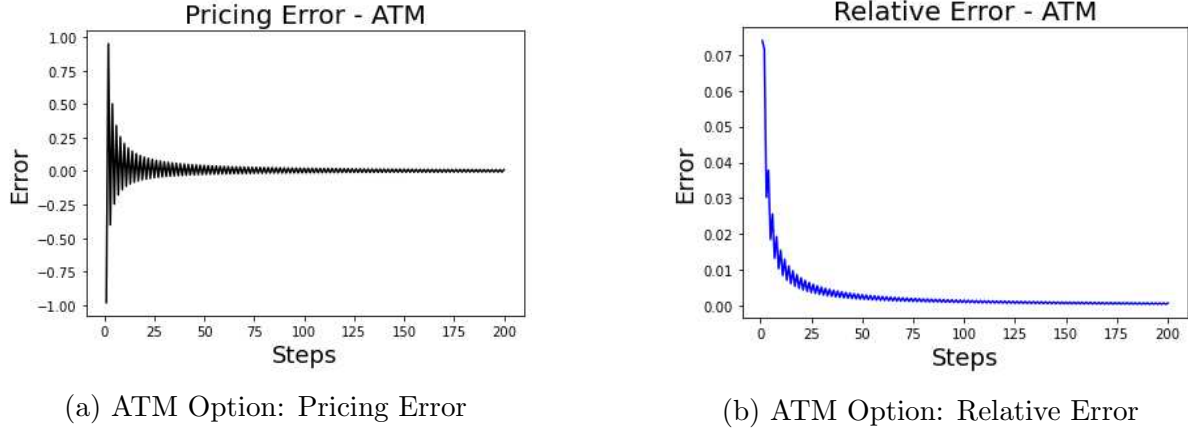


Figure 5: Convergence case: ATM Option

Other Results :

$$l_{\infty} = 0.9810187906522678$$

$$l_1 = 9.076799478458607$$

Experiment 2 - OTM Call

Here we consider the convergence when the option is OTM (i.e. using $X_1 > S_0$).

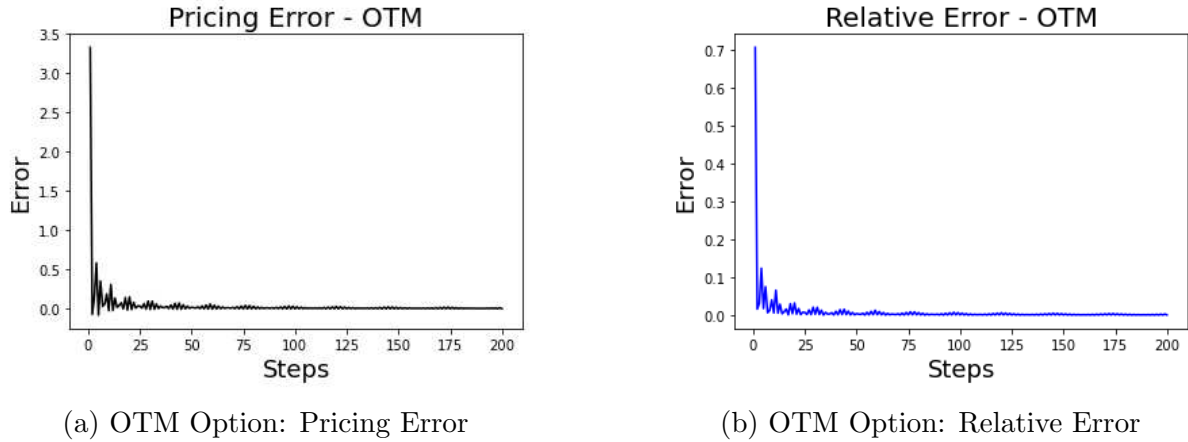


Figure 6: Convergence case: OTM Option

Other Results :

$$l_{\infty} = 3.330615414674594$$

$$l_1 = 8.166425643344734$$

Experiment 3 - ITM Call

Here we consider the convergence when the option is ITM (i.e. using $X_2 < S_0$).

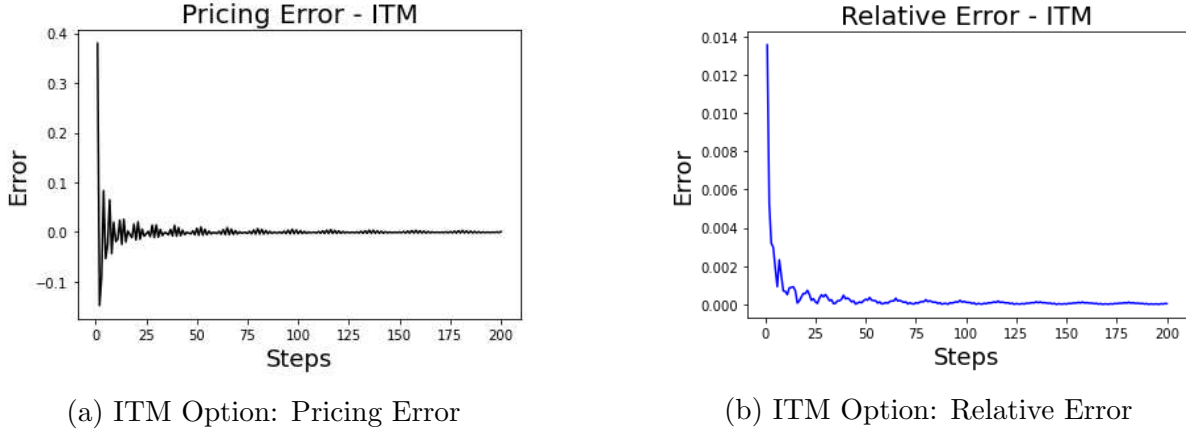


Figure 7: Convergence case: ITM Option

Other Results :

$$l_{\infty} = 0.3796562085203732$$

$$l_1 = 1.6305269699099512$$

What we see immediately is the common distinct saw-tooth pattern in the error plots as the CRR converges to the BSM. We see that the oscillation is much more pronounced when the option is ATM (or close to it, as we will see later in the paper). In all three cases the convergence oscillates up to $n = 50$ and starts to smooth out and settle up to $n = 100$. This is most often the optimal range of n observed in literature. However it is clear that beyond about 50 steps, the answer is close enough for most commercial purposes.

What is also noteworthy is that by looking at the l_{∞} -error and l_1 -error of each case we see that the l_{∞} -error is by far the greatest when the call is OTM and the smallest when ITM, however the l_1 -error is the greatest when the Call is ATM.

2.4 Why the CCR model is useful?

The question one might have is, if the CRR model requires as many steps as it does to approximate a model that already provides a better answer, what is the use?

2.4.1 The Advantage of the Binomial Approach

The key drawback of the Black-Scholes Model is that, it can only price European-style options. This means that options such as American, Bermudan and some path-dependant options like Asian and Barrier options cannot be priced using the Black-Scholes model. It cannot even take into account discrete dividends during the life of the underlying. This is where the CRR model has the advantage. Since the model simulates the path of the underlying at discrete intervals, it can take

into account aspects such as early exercise, discrete dividends, and path-dependencies. This makes the binomial model in general a powerful tool that is less complex and computationally intensive than its alternative the Monte Carlo simulation. This will be explored in a later paper in more detail. For now, we look at how the CRR model and the BSM can complement each other.

2.4.2 How the CRR and BSM model could complement each other

Let us consider a technique suggested by (Hull & White, 1988) called the Control Variate Technique. This technique can help improve the pricing of American Options for example. How this would work is, we would use the same tree we have constructed to price the European option and the American option, using the most optimal step sizes. This is represented by V_E^{CRR} and V_A^{CRR} respectively. From there, we calculate the value of the same European option using the Black-Scholes Model represented by V_E^{BSM} . We then simply assume that the error in pricing the European Option using the tree model is the same as the error for the American Option. The value for V_A^{BSM} (which cannot be calculated using the formula) can then be approximated by the following formula:

$$V_A^{BSM} = V_A^{CRR} + (V_E^{BSM} - V_E^{CRR})$$

3 Other models

There are many different variations of the Binomial Model (the CRR being the first). We will briefly look at 3 other models and summarise the rest found in literature.

3.1 Jarrow-Rudd Model (JR)

In order to solve the equations (2), (3), (Jarrow & Rudd, 1982) propose the following extra condition:

$$p = \frac{1}{2} \quad (16)$$

The solution to (2), (3) and (16) can be shown to be:

$$u = e^{\left(r - \frac{\sigma^2}{2}\right)\delta t + \sigma\sqrt{\delta t}} \quad (17)$$

$$d = e^{\left(r - \frac{\sigma^2}{2}\right)\delta t - \sigma\sqrt{\delta t}} \quad (18)$$

3.2 Tian Model

For this model, proposed by (Tian, 1999), the extra condition is:

$$pu^3 + (1 - p)d^3 = e^{3r\delta t + 3\sigma^2\delta t}. \quad (19)$$

The solution to (2), (3) and (19) can be shown to be:

$$u = \frac{RQ}{2} \left(Q + 1 + \sqrt{Q + 2Q - 3} \right) \quad (20)$$

$$d = \frac{RQ}{2} \left(Q + 1 + \sqrt{Q + 2Q - 3} \right) \quad (21)$$

$$p = \frac{1}{2} - \frac{3}{4}\sigma\sqrt{\delta t} \quad (22)$$

Where R is as defined before and $Q = e^{\sigma^2\delta t}$.

(Also note here that $ud \neq 1$. This means that the binomial tree loses its symmetry).

3.3 Summary of other models

Other models found in literature include:

- Rendleman-Bartter Model
- Chriss Model
- Wilmott1 and Wilmott2
- Jabbour-Kramin-Young
- Avellaneda and Laurence

These models go far beyond the scope of this paper, and in the results to follow it is clear why it isn't necessary to consider each in depth for the purposes of this paper.

3.4 Comparison

Instead of recreating each of the models discussed in this section and running multiple tests, for the sake of brevity, I consider the findings of similar comparison by (Chance, 2008).

The finance literature has revealed no fewer than 11 alternative versions of the binomial option pricing model for options on log-normally distributed assets. These models are derived under a variety of assumptions and in some cases require information that is ordinarily unnecessary to value options. This paper provides a review and synthesis of these models, showing their commonalities and differences and demonstrating how 11 diverse models all produce the same result in the limit. Some of the models admit arbitrage with a finite number of time steps and some fail to capture the correct volatility. This paper also examines the convergence properties of each model and finds that none exhibit consistently superior performance over the others.

The above however does not imply that the Binomial Model cannot be improved upon by means of other techniques as we will see in the following section.

4 Improving Convergence

4.1 Accelerated Binomial Method

In (Breen, 1991), the author describes the application of a convergence acceleration technique to the binomial option pricing model. The resulting model, termed the accelerated binomial option pricing model (ABOPM), follows from work done by (Geske & Johnson, 1984) (GJ), who proposed a model for the value of the American put. The ABOPM is accurate and faster than the conventional binomial model. It is applicable to a wide range of option pricing problems.

Both GJ and ABOPM is based on the Richardson extrapolation. In numerical analysis, Richardson extrapolation is a sequence acceleration method, used to improve the rate of convergence of a sequence of estimates of some value $V^* = \lim_{h \rightarrow 0} V(h)$. In essence, given the value of $V(h)$ for several values of h , we can estimate A^* by extrapolating the estimates to $h = 0$. It was introduced in the early 20th century by (Richardson, 1911). In the words of (Birkhoff & Rota, 1978), "its usefulness for practical computations can hardly be overestimated."

Consider the sequence of functions:

$$P_a(S) = \sum_j^{N-a} \binom{N-a}{j} \left(\frac{p}{q}\right)^j \left(\frac{(1-p)}{q}\right)^{N-a-j} V_{j,N-a} \quad (23)$$

where V is defined as earlier, converges to $P_N(S) = V_{0,0}$ for any binomial model with N periods. Convergence is uniform from below and occurs at $P_m(S)$, where m is the earliest period in the model for which $V_{0,m}$ takes its immediate exercise value rather than its holding value. m will always be less than N unless the put should be exercised immediately.

In other words, for a binomial option pricing model with fixed N , the value of the option is the limit of the sequence $P_a(S)$. $P_a(S)$ defines a sequence of option values with an increasing number of exercise opportunities. Thus, $P_a(S)$ is a European option, where exercise is only allowed time $t = N$. $P_1(S)$ is the value of an option if exercise is only allowed at period N and period $N - 1$; and so on. This sequence converges to $P_N(S)$, the true value of the option.

Consider now the related sequence, $P'_n(S)$, or $P'(n)$ for short, where $P'(1) = P_0(S)$; $P'(2) =$ binomial option value permitting exercise at N and $N/2$ only; $P'(3) =$ binomial option value permitting exercise at N , $2N/3$, and $N/3$ only. Again, this sequence converges to the option's value, $P_n(S)$ from below; that is, $P_n(S)$ is the limit of the sequence $P'(n)$ as $n \rightarrow N$.

To apply the Richardson extrapolation technique to the binomial, the author proceeds by analogy with GJ's exposition. They claim the parallels between the sequences P and P' are clear in that the number of exercise opportunities increases as we move down the sequence. They then apply GJ's Formula to the terms $P'(n)$ for $n = 1, 2, 3$. **The value of the option is then given as $\max(P', X - S)$.**

The author then tested the Binomial Method as proposed by CR as well as GJ against the ABOPM. Results where as follows:

All three sets of values agree very closely. The largest error in the AB method is of the order of one and a half cents compared with the B values. It is clear from their results that the GJ was more accurate. The key result here is the amount of computational reduction ABOPM method

TABLE 1
Values of American Put Option Using Binomial, Geske-Johnson, and Accelerated Binomial
Methods ($S = \$40$; $r = 1.05$)

X	σ	T	Binomial	Geske-Johnson	Accelerated Binomial
35.00	0.2000	0.08330	0.01000	0.006200	0.006000
35.00	0.2000	0.33330	0.20000	0.199900	0.198900
35.00	0.2000	0.58330	0.43000	0.432100	0.433800
40.00	0.2000	0.08330	0.85000	0.852800	0.851200
40.00	0.2000	0.33330	1.58000	1.580700	1.574000
40.00	0.2000	0.58330	1.99000	1.990500	1.984000
45.00	0.2000	0.08330	5.00000	4.998500	5.000000
45.00	0.2000	0.33330	5.09000	5.095100	5.102000
45.00	0.2000	0.58330	5.27000	5.271900	5.285000
35.00	0.3000	0.08330	0.08000	0.077400	0.077400
35.00	0.3000	0.33330	0.70000	0.696900	0.698500
35.00	0.3000	0.58330	1.22000	1.219400	1.224000
40.00	0.3000	0.08330	1.31000	1.310000	1.309000
40.00	0.3000	0.33330	2.48000	2.481700	2.476000
40.00	0.3000	0.58330	3.17000	3.173300	3.159000
45.00	0.3000	0.08330	5.06000	5.059900	5.063000
45.00	0.3000	0.33330	5.71000	5.701200	5.698000
45.00	0.3000	0.58330	6.24000	6.236500	6.239000
35.00	0.4000	0.08330	0.25000	0.246600	0.245000
35.00	0.4000	0.33330	1.35000	1.345000	1.350000
35.00	0.4000	0.58330	2.16000	2.156800	2.159000
40.00	0.4000	0.08330	1.77000	1.767900	1.766000
40.00	0.4000	0.33330	3.38000	3.363200	3.383000
40.00	0.4000	0.58330	4.35000	4.355600	4.339000
45.00	0.4000	0.08330	5.29000	5.285500	5.287000
45.00	0.4000	0.33330	6.51000	6.509300	6.505000
45.00	0.4000	0.58330	7.39000	7.383100	7.382000

For the binomial and accelerated binomial, $N = 150$. Geske-Johnson value is based on four-point extrapolation; Accelerated Binomial value is based on three-point extrapolation. Geske-Johnson values are from Geske and Johnson ((1984), p. 1519). Binomial values are from Cox and Rubinstein ((1985), p. 248).

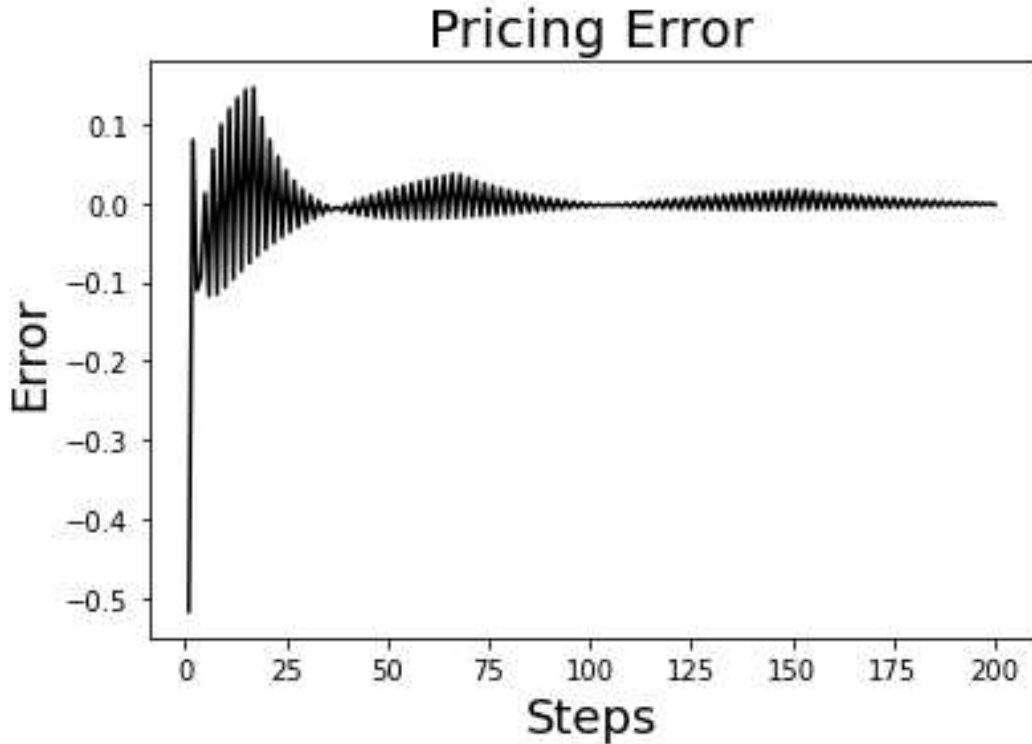
Figure 8: Results of the testing the GJ, CRR and ABOPM using 150 period model

brings. The results of the CRR method requires the calculation of $(N + l)/2$ node values, and for $N = 150$ that equates to 22801. The ABOPM model, on the other hand, requires only $4N + 10$ calculations, which is 610 for a $N = 150$.

At that point, the only previous attempts to investigate the application of convergence acceleration techniques to the binomial option pricing model where by Omberg (1987). The approach differed in that whereas ABOPM had shown how to accelerate the convergence of a particular binomial model with fixed N , Omberg sought to find a means by which to accelerate the convergence of a sequence of binomial option models with increasing N .

4.2 Odd/Even Step Method

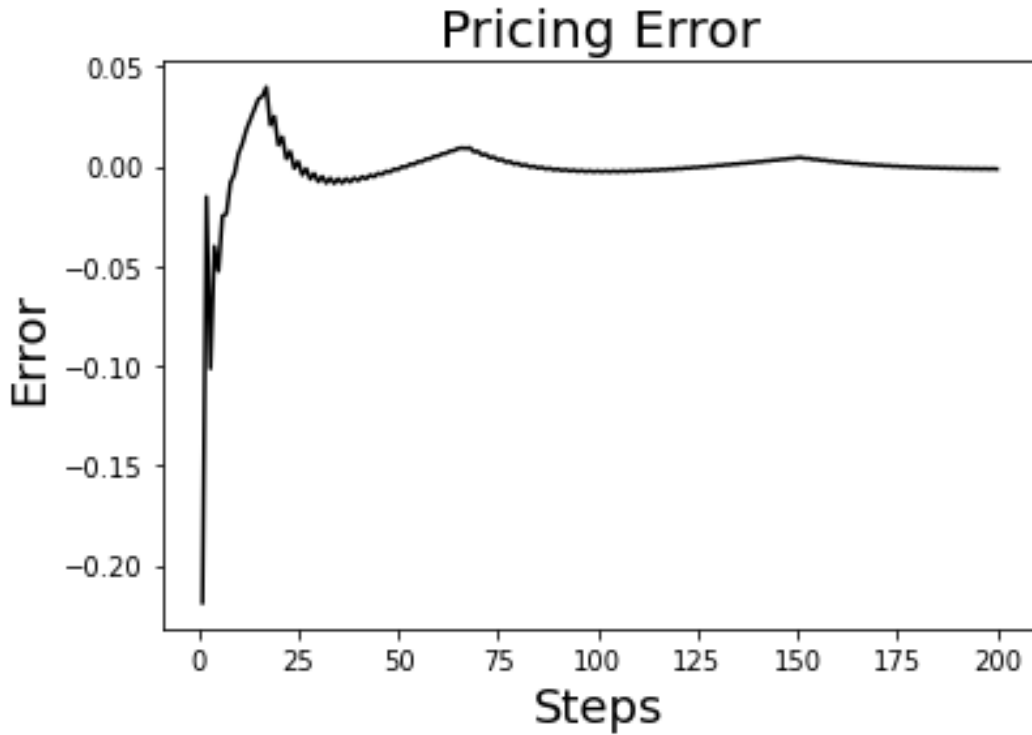
To illustrate what this method proposes, let's look at a clear illustration of the issue. We price a European Call the same as before but with a slight that the call is slightly in ITM. $S = 100$; $X = 110$; $r = 10\%$; $\sigma = 20\%$; $t = 1$ year



The solid appearance of the oscillation of convergence comes about because the answers obtained change more sharply in going from an odd number of steps to an even number than they do between successive odd or even numbers of steps. When the option price is plotted against the number of binomial steps, the result therefore zig-zags between the envelopes made up of odd and even numbers of steps.

The reasoning behind this is intuitively apparent: as n increases from 5 to 6, the way in which the binomial distribution is “fitted” to the normal distribution changes radically. For $n = 5$, the binomial distribution has two equal maximum probability values while for $n = 6$, there is only a single maximum probability; yet when n goes from 6 to 8, the only change is that a couple of extra bars are squeezed in, giving a slightly better approximation to the normal curve. One would therefore expect smooth transitions for the sequences $n = \dots 5, 7, 9, \dots$ and $n = \dots 6, 8, 10, \dots$ but jumps when going from odd to even to odd. (James, 2003)

In most circumstances, the answer obtained for n steps is improved on by taking the average of the answers of n steps and $(n + 1)$ steps. Even when the average of successive steps is taken, the value oscillates, with decreasing amplitude, around the analytical answer.



5 Conclusion and Future Research

From this paper it should be clear that the CRR model and the binomial pricing framework as a whole had a huge impact on option valuation when it was first established. There are some advantages of the Binomial model over the Black-Scholes framework in some cases as discussed, and in future work I will look at some of these cases in more depth. There are also a large variety of methods and extensions to the model that have been shown to increase accuracy and convergence. However, when you consider that institutions are running trades on option that reach far above the millions, it is clear that even very small errors will still have a significant impact on traders P&L statements. Fortunately there are much more accurate and complex methods to price options these days, such as Monte-Carlo simulations. In future work I would like to look at some of these methods as well.

References

- Birkhoff, G., & Rota, G.-C. (1978). *Ordinary differential equations (3rd ed.)*. John Wiley and sons.
- Black, F., & Scholes, M. (1973). The pricing of options and corporate liabilities. *Journal of political economy*, 81(3), 637–654.
- Breen, R. (1991). The accelerated binomial option pricing model. *The Journal of Financial and Quantitative Analysis*, Vol. 26(2), 153-164.
- Chance, D. M. (2008). A synthesis of binomial option pricing models for lognormally distributed assets.
- Cox, J. C., Ross, S. A., & Rubinstein, M. (1979). Option pricing: A simplified approach. *Journal of Financial Economics*, 7, 229-263.
- Geske, R., & Johnson, H. E. (1984). The american put option valued analytically. *The Journal of Finance*, 39(5), 1511–1524.
- Hull, J. (2009). *Options, futures, and other derivatives* (7th ed.). Pearson Prentice Hall.
- Hull, J., & White, A. (1988). The use of the control variate technique in option pricing. *Journal of Financial and Quantitative analysis*, 23(3), 237-251.
- James, P. (2003). *Option theory*. John Wiley & Sons.
- Jarrow, R., & Rudd, A. (1982). Approximate option valuation for arbitrary stochastic processes. *Journal of Financial Economics*, 10(3), 347 - 369.
- Merton, R. C. (1974). On the pricing of corporate debt: The risk structure of interest rates. *The Journal of finance*, 29(2), 449–470.
- Rendleman Jr, R. J., & Bartter, B. J. (1979). Two-state option pricing. *The Journal of Finance*, 34(5), 1093–1110.
- Richardson, L. F. (1911). Ix. the approximate arithmetical solution by finite differences of physical problems involving differential equations, with an application to the stresses in a masonry dam. *Philosophical Transactions of the Royal Society of London. Series A, Containing Papers of a Mathematical or Physical Character*, 210(459-470), 307–357.
- Thickstun, J. (2016). *Binomial option pricing and black-scholes* [Research Report].
- Tian, Y. S. (1999). A flexible binomial option pricing model. *Journal of Futures Markets: Futures, Options, and Other Derivative Products*, 19(7), 817–843.
- Wrigglesworth, A. (2018). *Numerical solution of the black-scholes equation using some explicit finite difference methods* (Unpublished master’s thesis). University of Pretoria.

Appendix A: Python Code for CRR model and the BSM model

```
# -*- coding: utf-8 -*-  
"""
```

```
Created on Mon Apr 27 10:35:35 2020
```

```
@author: alanw  
"""
```

```
import numpy as np  
from scipy.stats import norm
```

```
#####
```

```
# CRR Model
```

```
#####
```

```
def CRR(Put_Call, n, S_0, X, rfr, vol, t, AMNEUR):
```

```
    deltaT = t/n
```

```
    u = np.exp(vol*np.sqrt(deltaT))
```

```
    d = 1./u
```

```
    R = np.exp(rfr*deltaT)
```

```
    p = (R-d)/(u-d)
```

```
    q = 1-p
```

```
    # simulating the underlying price paths
```

```
    S = np.zeros((n+1,n+1))
```

```
    S[0,0] = S_0
```

```
    for i in range(1,n+1):
```

```
        S[i,0] = S[i-1,0]*u
```

```
        for j in range(1,i+1):
```

```
            S[i,j] = S[i-1,j-1]*d
```

```
    # option value at final node
```

```
    V = np.zeros((n+1,n+1)) #  $V[i,j]$  is the option value at node  $(i,j)$ 
```

```
    for j in range(n+1):
```

```
        if Put_Call=="C":
```

```
            V[n,j] = max(0, S[n,j]-X)
```

```
        elif Put_Call=="P":
```

```
            V[n,j] = max(0, X-S[n,j])
```

```
    # European Option: backward induction to the option price  $V[0,0]$ 
```

```
    if AMNEUR == "E":
```

```
        for i in range(n-1,-1,-1):
```

```
            for j in range(i+1):
```

```
                V[i,j] = max(0, 1/R*(p*V[i+1,j]+q*V[i+1,j+1]))
```

```
            opt_price = V[0,0]
```

```
    # American Option: backward induction to the option price  $V[0,0]$ 
```

```
    elif AMNEUR == "A":
```

```
        for i in range(n-1,-1,-1):
```

```
            for j in range(i+1):
```

```
                if Put_Call=="P":
```

```
                    V[i,j] = max(0, X-S[i,j],  
                                1/R*(p*V[i+1,j]+q*V[i+1,j+1]))
```

```
                elif Put_Call=="C":
```

```
                    V[i,j] = max(0, S[i,j]-X,  
                                1/R*(p*V[i+1,j]+q*V[i+1,j+1]))
```

```
            opt_price = V[0,0]
```

```
    return opt_price
```

```
#####
```

```
# BSM Model
```

```
#####
```

```

def BSM(Put_Call, S_0, X, rfr, vol, t):
    d_one = (np.log(S_0/X)+(rfr+vol**2/2)*t)/(vol*np.sqrt(t))
    d_two = d_one - vol*np.sqrt(t)

    if Put_Call == "C":
        opt_price = S_0*norm.cdf(d_one) - X*np.exp(-rfr*t)*norm.cdf(d_two)
    elif Put_Call == "P":
        opt_price = X*np.exp(-rfr*t)*norm.cdf(-d_two) - S_0*norm.cdf(-d_one)

    return opt_price

#####
# Pricing Error
#####
def Price_Error(Put_Call, n, S_0, X, rfr, vol, t):
    AMNEUR = "E"
    error = BSM(Put_Call, S_0, X, rfr, vol, t)-CRR(Put_Call, n, S_0, X, rfr,
                                                    vol, t, AMNEUR)

    return error

#####
#Avg Even/Odd
#####
def Avg_Even_Odd_Error(Put_Call, n, S_0, X, rfr, vol, t):
    AMNEUR = "E"
    error = BSM(Put_Call, S_0, X, rfr, vol, t)-
    ((CRR(Put_Call, n, S_0, X, rfr, vol, t, AMNEUR)+
      CRR(Put_Call, n+1, S_0, X, rfr, vol, t, AMNEUR))/2)

    return error

```

Appendix B: Other non-binomial numerical methods

Trinomial Models

Suppose the asset price is S_0 at time $t = 0$. After one period δt the asset price can be uS_0 (up-state) with probability p_u , mS_0 (middle-state) with probability p_m , and dS_0 (down-state) with probability p_d . The tree now looks something like this:

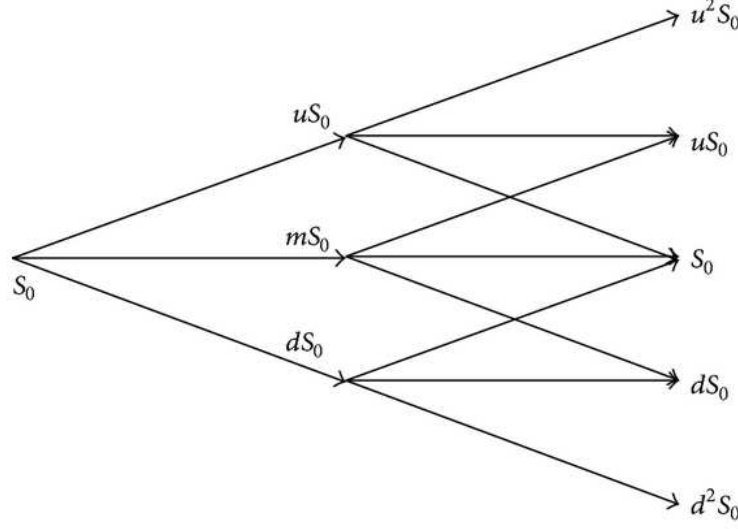


Figure 9: Two-step Trinomial Lattice Tree

Then the risk neutral one period option pricing formula is:

$$V(S_t, t) = \frac{1}{R} (p_u V(uS_t, t + \delta t) + p_m V(mS_t, t + \delta t) + p_d V(dS_t, t + \delta t)) \quad (24)$$

The trinomial tree approach is equivalent to the explicit finite difference method. By equating the first and second moments of the asset price $S_{\delta t}$ in both continuous and discrete models, we obtain the following equations:

$$p_u + p_m + p_d = 1 \quad (25)$$

$$p_u u + p_m m + p_d d = e^{r\delta t} \quad (26)$$

$$p_u u^2 + p_m m^2 + p_d d^2 = e^{(2r + \sigma^2)\delta t} \quad (27)$$

There are three equations and six variables u, m, d and p_u, p_m, p_d . We need three more equations to uniquely determine these variables.

Boyle Model

The three additional conditions are

$$m = 1, ud = 1, u = e^{\lambda\sigma\sqrt{\delta t}} \quad (28)$$

where λ is a free parameter. -insert probabilities-

Hull-White

This is the same as the Boyle model where the choice of λ is $\lambda = \sqrt{3}$. -insert probabilities-

Finite Difference Methods

Finite difference methods are numerical methods that attempt to solve partial differential equations. If they are applied correctly, they are capable of producing accurate numerical solutions.

The underlying idea is quite simple. The continuous differential equation is replaced with a discrete approximation. This is done by replacing each of the derivative terms with finite difference approximations. So the numerical solution is only known at a finite number of points. The number of points can be chosen and, in general, increasing the number of points improves the accuracy of the numerical solution, however computational time can become too large. Consider the function $u(x, t)$, where x is the spatial variable and t the

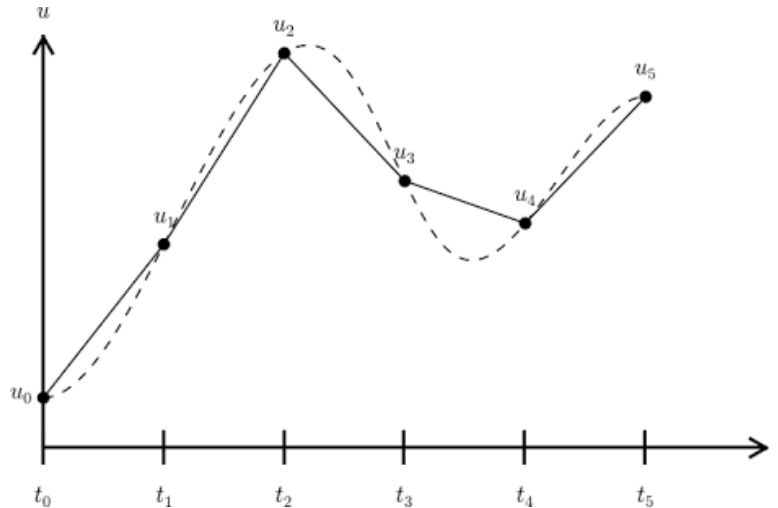


Figure 10: finite difference approximation of $u(x, t)$

temporal variable. We start by dividing the x -axis into equally spaced nodes, with distance Δx . Similarly we divide the t -axis into equally spaced nodes, with distance Δt

This breaks the (x, t) -plane into a mesh or grid. The most popular finite difference methods that are used include:

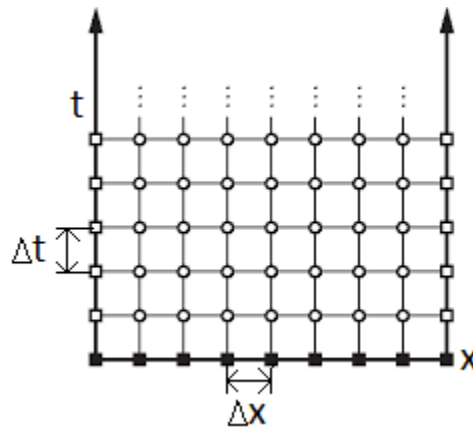


Figure 11: The finite grid/mesh of the (x, t) -plane

- Explicit Methods
- Implicit Methods
- Hybrid Methods (e.g. Crank-Nicolson)

In (Wrigglesworth, 2018) I delve deeper into some specific explicit methods applied to a transformed Black-Scholes equation.